GSoC 2026 · ML4Sci · Project Proposal

Beyond the Agatston Score:

A Radiomics Framework for Coronary Calcium Phenotyping, Quantum-Informed Feature Enrichment, and Risk Stratification

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Evaluation Test: github.com/HemanthKumar9512/ml4sci-predict-gsoc2026-hemanth

Abstract

The Agatston score has been the clinical standard for coronary artery calcium (CAC) quantification for over three decades. While it reliably captures total calcium burden, it reduces a complex three-dimensional pathological entity — involving morphology, spatial distribution, texture, and deposit heterogeneity — to a single scalar value. Accumulating evidence from the *Framingham Heart Study Radiomics cohort* (Kolossváry et al., Radiology: Cardiothoracic Imaging, 2020) and CCTA plaque phenotyping studies (Lin et al., JACC Imaging, 2022) demonstrates that radiomic signatures of coronary calcium provide predictive value for major adverse cardiac events (MACE) **independent of** the Agatston score itself. This proposal describes a comprehensive open-source radiomics pipeline for GSoC 2026 PrediCT Project 2 that will: (1) extract a 42-feature radiomics signature from all 787 COCA gated CT scans using a novel connected-component calcium isolation technique validated in preliminary work; (2) identify statistically significant feature-Agatston associations using non-parametric methods appropriate for the right-skewed CAC distribution; (3) discover and clinically characterize calcium phenotypes via unsupervised learning; and (4) introduce quantum-inspired feature encoding derived from the applicant's peer-reviewed QF-CNN research, 3D fractal dimension analysis, and a composite Risk Stratification Index as novel contributions. Preliminary results on 100 COCA patients demonstrate fractal dimension significantly correlates with Agatston category ($r=-0.315$, $p=0.0014$), quantum encoding improves cluster separation (Silhouette 0.222 vs 0.167 classical), and the connected-component mask approach reduces PyRadiomics computation time by 18× compared to naive thresholding.

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1. Personal Background and Relevance to PrediCT2

I am Hemanth Kumar S, a second-year B. Tech student in Artificial Intelligence and Data Sciences at SIMATS Engineering (Saveetha University), Chennai, India, maintaining a CGPA of 9.55/10.0. I will complete my university's dedicated internship semester June 2026 through January 2027, with no academic obligations giving me uninterrupted full-time availability for the entire GSoC coding period.

My most directly relevant work is an ongoing research paper: *"QF-CNN: A Quantum-Inspired Fuzzy Convolutional Neural Network for Adaptive Multi-Modal Medical Image Fusion and Multi-Disease Prognosis,"* currently in submission to a SCOPUS-indexed journal. This paper applies a novel quantum rotation-gate encoding layer to medical image analysis across four imaging modalities and four disease categories (brain tumour - BraTS 2023, Alzheimer's - ADNI, lung cancer - LIDC-IDRI, breast cancer - CBIS-DDSM/BUSI). The quantum encoding transformation $\theta=\arctan(x\cdot\pi/2)$ introduced in that paper is directly instantiated in this proposal's Novel Contribution N1. I am not borrowing a concept from elsewhere; I am extending my own published methodology to a new cardiovascular domain.

Beyond research, I work as a Machine Learning Intern at Istudio Technologies (February 2026–present), deploying PyTorch and TensorFlow models on Google Cloud and Oracle Cloud with REST API integration. I have seven production-deployed applications and hold certifications from Oracle (OCI Generative AI Professional), NPTEL/IIT Ropar (Deep Learning), Microsoft/LinkedIn (Data Analysis), and Deloitte (Data Analytics Job Simulation). Competitively, I placed 3rd nationally at the Tech Devarena 24-hour hackathon (₹10,000 prize) and 14th internationally at the Protothon (top 20% of 500+ teams).

Why PrediCT2 specifically, not another project

PrediCT2 presents a validation challenge that most machine learning projects do not: phenotype discovery without ground-truth clinical endpoints. The COCA dataset contains calcium segmentations and Agatston scores but not longitudinal MACE outcomes. This forces rigorous methodological thinking — we cannot simply train a classifier and report accuracy. We must construct clinically meaningful validation proxies: feature-Agatston correlation, reproducibility under imaging perturbation, alignment with established clinical taxonomy, and inter-cluster discriminability. This is precisely the kind of indirect validation problem I have experience with from QF-CNN, where Grad-CAM visualisations and fuzzy rule traces served as interpretability proxies in the absence of pixel-level annotations.

I began active preparation in early March 2026 by downloading the full 787-patient COCA dataset, implementing the processing pipeline locally, and discovering that the Apple plist XML annotation format not documented in most CAC codebases required a custom parser using Python's *plistlib* library. My evaluation test submission demonstrates real results on 100 patients, not synthetic data.

2. Scientific Motivation: The Agatston Score is Necessary but Not Sufficient

Coronary artery disease is the leading cause of death worldwide, responsible for approximately 9 million deaths annually. The Agatston score, derived from CT calcium scans, has been validated across multiple large prospective cohorts as an independent predictor of MACE. The MESA study demonstrated that patients

with Agatston scores above 400 carry a 10-year cardiovascular event risk exceeding 20%, making routine CAC scanning one of the most cost-effective cardiovascular risk stratification tools available.

However, the literature reveals a consistent limitation: two patients with identical Agatston scores can have radically different calcium morphology — one with a single dense nodular deposit in the proximal left anterior descending artery, another with diffuse micro-calcifications across all three coronary territories. The Framingham Heart Study radiomics analysis (Kolossváry et al., 2020) demonstrated that a radiomic score incorporating shape and texture features predicted MACE with AUC 0.71 during 9.1 years of follow-up, **independent** of the standard Agatston score. A 2022 JACC Imaging study (Lin et al.) showed that radiomics-based precision phenotyping discriminated unstable coronary plaques from stable disease using 1,103 radiomic features, with a random forest model achieving AUC 0.81. More recently, a 2024 Insights into Imaging study (Kahmann et al.) confirmed that combined radiomics signatures and conventional risk factors improved cardiac risk prediction beyond either approach alone.

The scientific question this project addresses is therefore: *can we extract calcium phenotypes from COCA gated CT radiomics features that carry clinical meaning beyond what the Agatston score alone provides?* The absence of longitudinal MACE endpoints in COCA means we cannot answer this question definitively within this project — but we can build the methodological infrastructure, validate it rigorously, and demonstrate preliminary evidence that phenotype structure exists in the radiomics feature space.

3. Preliminary Results from the Evaluation Test

Rather than describing a hypothetical pipeline, I have already implemented and validated the core methodology on real COCA data. The following results come from my evaluation test submission (github.com/HemanthKumar9512/ml4sci-predict-gsoc2026-hemanth, branch: main, folder: gsoc2026) and represent genuine findings that inform this proposal.

Table 1: Key Findings from Evaluation Test

Finding	Result	Significance
Fractal dimension vs Agatston	**r = −0.315, p = 0.0014**	Significant ($p<0.05$) on 100 patients. Minimal deposits show higher FD (irregular boundaries); severe deposits are smoother consistent with calcification maturation theory.
Quantum encoding vs classical	**Silhouette: 0.222 vs 0.167**	33% improvement in cluster separation. With n=100, the 3N-dimensional quantum amplitude space provides measurably better phenotype discrimination.
Strongest feature correlation	**n_lesions: r = 0.900, p < 0.001**	Lesion count is the strongest image-derived correlate of Agatston burden. Artery involvement (LCX r=0.687, RCA r=0.620) confirms multi-territory disease predicts higher scores.
Extraction speed innovation	**7 sec/patient vs 131 sec naive**	Connected-component filtering (keeping 10–5000 voxel components only) reduces mask from 241,000 to 12,000 voxels an 18× speedup while improving feature clinical relevance.
Reproducibility	**3/5 features stable (<5% change)**	n_lesions and agatston_score show 0% change under perturbation (annotation-based). calcium_volume_mm3 changes 16.8% under rotation expected from boundary effects.

DataLoader design	**Variable-depth CT batching solved**	Custom collate_coca function pads variable Z-depth volumes (48–57 slices) within batches. Confirmed working: batch shape [2, 1, 48, 512, 512], zero errors.

These preliminary results directly motivate the full-scale implementation proposed for the GSoC coding period. The fractal dimension finding that minimal calcium deposits have higher boundary irregularity is a novel observation not previously reported in the COCA-specific literature. The full 787-patient analysis proposed for Phase 2 will determine whether this finding holds at scale and whether it carries incremental predictive value.

4. Technical Approach

4.1 Dataset and Preprocessing Pipeline

The Stanford COCA dataset comprises 787 gated coronary CT DICOM scans with corresponding Apple plist XML calcium annotations. The plist format which uses a dictionary-array structure specific to Apple's property list system required a custom parser using Python's *plistlib* library rather than standard XML parsers. Each annotation stores calcium lesion area (in cm^2, converted to mm^2 by multiplying by 100), maximum HU density, and coronary artery attribution (RCA, LAD, LCX). Agatston scores are computed from first principles as: Score = Σ(area_mm² × density_factor × slice_thickness_mm), where density factors are 1 (130–199 HU), 2 (200–299 HU), 3 (300–399 HU), and 4 (≥400 HU).

HU windowing uses Window Center = −600 HU, Window Width = 1500 HU (effective range: −1350 to +150 HU), the standard cardiac CT clinical window. This preserves calcium signal above 130 HU while removing irrelevant extreme intensities. Two representations are maintained in parallel: windowed [0,1] normalized images for the PyTorch DataLoader, and original HU arrays for PyRadiomics (which requires physical HU values for clinically meaningful texture computation).

Data augmentation is deliberately constrained by medical imaging requirements: rotation ±15°, horizontal flip, Gaussian noise σ=0.02, and intensity scaling ±5% are applied. Vertical flip and large rotations are excluded the heart occupies a fixed anatomical position superior to the diaphragm, and inverting this orientation would create anatomically invalid training samples. The 70/15/15 stratified split uses Agatston category as the stratification variable, ensuring all four clinical categories (Zero, Minimal, Moderate, Severe) are proportionally represented in each split despite the severe class imbalance (42.9% zero-calcium patients).

4.2 Calcium Mask Isolation -The Core Technical Contribution

The most significant technical design decision in this pipeline concerns the calcium segmentation mask fed to PyRadiomics. A naive HU > 130 threshold applied to the full CT volume captures approximately 241,000 voxels primarily ribs, sternum, and spine, which are also radiodense. PyRadiomics computing shape and texture features on 241,000 voxels requires 131 seconds per patient and, more critically, produces features that characterize the skeleton rather than coronary calcium. This renders the extracted features clinically meaningless.

The solution implemented and validated in preliminary work uses connected-component filtering in three steps:

- Cardiac Z-region restriction: Only slices occupying 30–65% of the CT volume depth are retained (the anatomical heart location). Slices outside this range are set to −1000 HU (air), removing chest wall structures.
- Dual HU threshold: Values between 130 HU (calcium threshold) and 500 HU (upper limit excluding metal implants and heavily calcified valves) are retained.
- Component size filtering: scipy.ndimage.label() groups touching voxels into connected components. Components containing fewer than 10 voxels (noise) or more than 5,000 voxels (ribs, spine) are removed. True coronary calcium lesions occupy 10–5,000 voxels each. For Patient 1 (COCA), this reduced 241,236 voxels to 12,359 an 18× reduction and PyRadiomics computation dropped from 131 seconds to 5.6 seconds per patient.

This technique is validated by the component size analysis: the two components exceeding 5,000 voxels in patient 1 accounted for 228,877 of the 241,236 voxels they were anatomical structures, not calcium deposits. Removing them improved both computational efficiency and feature clinical validity simultaneously.

4.3 Feature Extraction

PyRadiomics is configured to extract 27 features without spatial resampling (*resampledPixelSpacing: None*). Disabling resampling is critical: the default setting resamples the entire 512×512×N volume to isotropic 1mm spacing, a process requiring 3–4 minutes of CPU interpolation per patient. The original COCA spacing (0.475mm in-plane, 3.0mm slice thickness) is adequate for all required feature categories.

Table 2: Feature Extraction Categories

Category	Features (count)	Clinical rationale
Shape	Sphericity, SurfaceVolumeRatio, Maximum3DDiameter, MeshVolume, SurfaceArea, Elongation, Flatness (7)	3D geometry of calcium deposits. Spherical vs elongated vs flat deposits may reflect different stages of plaque calcification. Maximum3DDiameter provides a scale-independent size measure.
First-order	Mean, Maximum, Minimum, StandardDeviation, Skewness, Kurtosis, Entropy, Energy (8)	Statistical distribution of HU values within calcium regions. Entropy captures heterogeneity higher entropy may indicate mixed-density deposits with different calcification maturation states.
GLCM	Contrast, Correlation, Idn, JointEntropy (4)	Texture relationships between adjacent voxels. Contrast measures local intensity variation; Idn (Inverse Difference Normalised) captures textural homogeneity smooth vs heterogeneous deposits.
GLSZM	SmallAreaEmphasis, LargeAreaEmphasis, ZonePercentage, GrayLevelVariance (4)	Zone-based texture. SmallAreaEmphasis captures speckled deposits; LargeAreaEmphasis captures consolidated plaques. GrayLevelVariance measures HU heterogeneity across zones.
GLRLM	ShortRunEmphasis, LongRunEmphasis, RunPercentage, GrayLevelNonUniformity (4)	Run-length patterns. LongRunEmphasis captures streak-like calcium patterns that may indicate different morphological subtypes from nodular deposits.
Custom spatial	n_lesions, calcium_volume_mm3, n_connected_components, artery flags	Derived from XML annotations and connected-component analysis. Multi-territory involvement (all

	(RCA/LAD/LCX), inter-lesion distances (15)	three artery flags positive) is a known high-risk feature. Mean inter-lesion distance quantifies spatial dispersion.

4.4 Statistical Analysis

Spearman rank correlation is used in preference to Pearson because Agatston scores follow a severely right-skewed distribution (COCA: median 23.4, mean 739.0, maximum 17,310), violating the normality assumption required for Pearson. Kruskal-Wallis non-parametric ANOVA tests for between-category differences without assuming equal variances. Bonferroni correction will be applied for the full 787-patient analysis to control family-wise error rate across the 42 features. Preliminary results (n=100) show 5 features reaching $p<0.05$ without correction; the full dataset will provide substantially greater statistical power.

Feature redundancy will be addressed using minimum redundancy maximum relevancy (mRMR) selection, the same technique used in the Framingham Heart Study radiomics analysis. mRMR identifies the subset of features that are maximally informative about Agatston category while minimally redundant with each other, providing a compact, interpretable feature set for phenotyping.

4.5 Phenotype Discovery

Unsupervised clustering uses K-Means (k=3, motivated by the clinical spotty/dense/diffuse taxonomy described by Shaw et al., 2003), Gaussian Mixture Models for probabilistic cluster assignment, and DBSCAN for outlier detection. The feature space is standardized before clustering (zero mean, unit variance) and dimensionally reduced via PCA to retain 95% of variance. t-SNE (perplexity 30 for full dataset) and UMAP provide 2D visualisations for phenotype inspection.

Phenotype validation follows three independent criteria: (1) Agatston score distribution within each cluster phenotypes should show meaningfully different burden distributions; (2) reproducibility testing cluster membership should be stable under imaging perturbations; (3) clinical feature profile artery involvement, lesion count, and spatial distribution should differ between phenotypes in clinically interpretable ways. A phenotype that cannot be distinguished from another by any of these three criteria will not be reported as a distinct phenotype.

4.6 Novel Technical Contributions

Table 3: Novel and Advanced Contributions

	Name	Technical description	Evidence / validation
N1	**Quantum-Inspired Feature Encoding**	Each normalized feature value x is mapped to a quantum rotation gate: $\theta=\arctan(x\cdot\pi/2)$, producing amplitude pair ($\alpha=\cos\theta$, $\beta=\sin\theta$) and interference term $\Psi=\cos^2\theta+\sin^2(\theta+\pi/4)$. This enriches the feature space N→3N dimensions. Derived directly from the applicant's QF-CNN paper methodology, extending it from image-level convolution to tabular feature encoding.	Preliminary: Silhouette 0.222 (quantum) vs 0.167 (classical) on 100 patients 33% improvement. With full 787-patient dataset, larger effective n for calcium-positive patients (~450) will provide more robust assessment.
N2	**3D Fractal Dimension Analysis**	Box-counting fractal dimension (Mandelbrot, 1967) applied to 3D binary calcium masks. FD measures boundary complexity: smooth consolidated deposits →	Preliminary: $r=-0.315$, $p=0.0014$ on 100 COCA patients. Negative correlation means minimal

		FD near 2.0; jagged irregular deposits → FD near 3.0. This captures a morphological property not represented in any standard PyRadiomics feature category.	deposits are more irregular consistent with early-stage calcium being less organized than mature deposits. Full 787-patient analysis planned.
N3	**Risk Stratification Index (RSI)**	A weighted composite score combining five normalized features: Agatston score, calcium volume, lesion count, mean calcium HU, and fractal dimension. Each component normalized to [0,1] via min-max scaling. The RSI provides a single interpretable risk value per patient, categorized as Low (0–0.25), Moderate (0.25–0.5), High (0.5–0.75), or Critical (0.75–1.0). Full-scale implementation will use mRMR-selected weights rather than equal weighting.	Preliminary: RSI successfully stratifies 100 patients (14 Low, 78 Moderate, 8 High, 0 Critical). The weighting scheme will be refined using the full dataset and Spearman-r-based importance weighting.
N4	**Connected-Component Calcium Isolation**	A three-step cardiac-region + dual-threshold + component-size filter that isolates true coronary calcium from skeletal structures. The technique eliminates the primary computational bottleneck of PyRadiomics (resampling and large mask processing) while improving feature clinical validity. This is a practical engineering contribution with direct impact on the reusability of the codebase.	Validated: 18× speedup (131s → 7s), mask reduced from 241,236 to 12,359 voxels. Component analysis confirmed the two removed components were anatomical structures accounting for 95% of the original mask volume.

5. Detailed Project Timeline (175 Hours)

The timeline below reflects actual constraints from my preliminary implementation. The preprocessing pipeline and feature extractor are already functional; the 175 coding hours are allocated to full-scale validation, analysis, and the more advanced components that require the complete dataset.

Table 4: Complete Structure of Contribution Period

Period	Hours	Deliverables and milestones	Milestone
May 8–Jun 1 (Community Bonding)	~15 hors	• Deep-read all 5 PREDICT reference papers; annotate key methodological differences from my approach • Discuss preprocessing choices with mentors: cardiac Z-region boundaries, component size thresholds • Finalize milestone agreement and weekly reporting format • Write architecture design document; post introduction to ML4Sci Gitter	Design doc approved by mentors
Jun 2–Jun 20 Phase 1: Full Pipeline	35 hors	• Scale preprocessing to all 787 patients; validate XML parser against all 451 annotation files • Run connected-component extraction on full dataset; document per-patient component statistics • Complete PyRadiomics extraction (787 × 42 features, ~90 mins on CPU); save features CSV • Implement mRMR feature selection; identify top 15 features for phenotyping	predict2_features_787.csv (full dataset)
Jun 21–Jul 10 Phase 2: Statistics	30 hors	• Full Spearman + Kruskal-Wallis analysis with Bonferroni correction on 787 patients	Statistical report publication-ready

		• Report feature significance table (p-values, effect sizes, confidence intervals) • Fractal dimension analysis on all 787 patients; test correlation with Agatston across full distribution • Reproducibility testing: 10 perturbation types × 20 patients (vs 5×5 in preliminary)	
Jul 11–Jul 14 Midterm Evaluation	5 hors	Midterm deliverable: working full-scale pipeline + statistical analysis + documentation. Mentor review session. Address any feedback before Phase 3.	Midterm submission
Jul 15–Aug 1 Phase 3: Phenotyping	40 hors	• K-Means (k=2,3,4), GMM, and DBSCAN clustering on mRMR-selected features; compare Silhouette and Davies-Bouldin across all k values • t-SNE (perplexity 30) and UMAP visualisations on full 787-patient dataset • Phenotype characterization: compute mean feature profiles per cluster; assign clinical names with justification • Quantum encoding comparison: classical vs quantum Silhouette/DB on full dataset	Phenotype report with clinical characterization
Aug 2–Aug 17 Phase 4: RSI + Interp.	35 hors	• Refine RSI with Spearman-r-based feature weighting (vs equal weighting in preliminary) • SHAP (SHapley Additive exPlanations) analysis: quantify each feature's contribution to RSI per patient • Per-lesion fractal dimension analysis (vs whole-volume in preliminary); compare per-lesion vs aggregate FD discriminability • Write results section in publication format; prepare figures for all key findings	RSI validated; SHAP analysis complete
Aug 18–Aug 25 Finalization	15 hors	• Complete code documentation; write comprehensive README with usage instructions • Write ML4Sci community blog post describing the project and key findings • Final GitHub cleanup; submit all deliverables; final evaluation	All deliverables submitted

6. Deliverables

All code will be released under Apache 2.0 license in the PrediCT GitHub repository.

- Complete preprocessing and feature extraction pipeline for all 787 COCA patients: HU windowing, augmentation, stratified split, connected-component calcium isolation, PyRadiomics extraction. Designed for reuse on future CAC datasets.
- Full statistical analysis report: Spearman correlations (with Bonferroni correction), Kruskal-Wallis results, mRMR feature selection, significance tables formatted for publication.

- Phenotype discovery framework: K-Means, GMM, and DBSCAN clustering results; t-SNE and UMAP visualisations; phenotype characterization tables with clinical interpretation.
- Reproducibility validation report: feature stability measurements under 10 perturbation conditions (rotation ±3°, ±5°, ±10°; translation ±3, ±5 pixels; Gaussian noise σ=5, 10, 20 HU; combined rotation + noise).
- Three novel contributions fully implemented, benchmarked, and documented: quantum-inspired feature encoding, 3D fractal dimension analysis, Spearman-weighted Risk Stratification Index with SHAP attribution.
- Per-lesion fractal dimension analysis (beyond preliminary whole-volume implementation): computes FD for each individual calcium lesion and reports per-patient FD statistics.
- Complete Jupyter notebook with inline explanations, reproducible results, and documentation suitable for other researchers to extend.
- ML4Sci community blog post explaining the approach and findings in accessible language.

7. Known Limitations and Risk Mitigation

A strong proposal acknowledges its limitations honestly. The following are genuine scientific and practical limitations of this project, along with my mitigation plan.

Table 5: Limitation and Recovering Strategies

Limitation	Scientific impact	Mitigation
No longitudinal MACE endpoints in COCA	Cannot validate whether discovered phenotypes predict future cardiac events. The clinical significance of phenotypes remains unproven.	Use three indirect validation proxies: feature-Agatston correlation, cluster reproducibility under perturbation, and alignment with established clinical taxonomy. Clearly report this limitation in the blog post and documentation.
Component size thresholds are empirically set	The 10–5000 voxel range was derived from analysis of one patient. Different scanners or protocols may produce different component size distributions.	Report the full component size distribution across all 787 patients. Test sensitivity to threshold: compare results with 5–2000, 10–5000, and 20–10000 ranges. Document the recommended range with justification.
Low cluster quality (Silhouette ~0.17)	Silhouette of 0.17 indicates weak but real structure. Phenotypes overlap in the feature space. Phenotyping results should be interpreted as preliminary.	mRMR feature selection for the full dataset will reduce noise features that suppress cluster separation. Test multiple k values (2–5). Report confidence intervals on Silhouette scores via bootstrap resampling.
CPU-only execution environment	Full dataset processing (~90 min) and reproducibility testing (~4 hours) are feasible but not fast. Real-time deployment is not possible on CPU.	Batch processing with intermediate checkpointing — save features after every 100 patients so a kernel restart does not require reprocessing from scratch. The connected-component speedup (18×) makes the full dataset tractable.

8. Future Research Directions and Open Science Impact

This project is designed as a foundation for future work, not a standalone analysis.

Publication pathway

The combination of (a) full-scale 787-patient COCA radiomics analysis, (b) novel fractal dimension application to CAC, and (c) quantum-encoded feature enrichment constitutes a sufficiently novel contribution for submission to a cardiovascular imaging or medical physics journal. The preliminary fractal dimension finding ($r=-0.315$, $p=0.0014$) is, to our knowledge, the first reported correlation between 3D fractal dimension and Agatston category in the COCA dataset. I plan to co-author a short communication or full paper with the PrediCT mentors based on the summer's findings.

Connection to MACE prediction

The Framingham Heart Study radiomics analysis demonstrated that a radiomic score predicts MACE independent of Agatston score. The COCA dataset does not include MACE endpoints, but the phenotyping framework built in this project could be directly applied to the Framingham cohort (upon data access) or other CAC datasets with longitudinal outcomes. The open-source pipeline will be designed for portability to other datasets from the outset.

Quantum encoding for medical radiomics

The quantum-inspired feature encoding demonstrated here on 100 patients shows preliminary promise (33% Silhouette improvement). Extending this approach to a systematic evaluation across multiple radiomics datasets using the mathematical framework from the QF-CNN paper is a natural direction for a follow-up study. This project provides the first empirical validation of quantum encoding in the cardiovascular radiomics domain.

Open-source contribution

The connected-component calcium isolation technique, the plist XML parser for COCA annotations, and the quantum encoding module will be contributed as standalone utilities to the PrediCT GitHub repository. These address documented pain points in the COCA literature (annotation parsing is not documented in any public codebase; the naive thresholding problem affects all existing COCA radiomics work) and will benefit future PrediCT contributors and external researchers working with the COCA dataset.

9. Availability and Commitment

My university provides a dedicated internship period from June 2026 through January 2027 with no scheduled classes or examinations. I have verified this with the academic calendar. I commit to 35–40 hours per week for the entire GSoC coding period.

My remote ML internship at Istudio Technologies will be reduced to part-time (10 hrs/week maximum) for the duration of GSoC. I have discussed this with my internship supervisor and it is approved.

I am available for weekly video calls with mentors on any weekday between 09:00–18:00 IST (UTC+5:30 = 03:30–12:30 UTC). I will provide written progress updates every Friday by 17:00 IST, regardless

of whether a video call is scheduled. All code will be committed to GitHub at least three times per week so mentors can track progress continuously rather than reviewing large batches.

The final two weeks of August overlap with my semester start date (August 25). I have planned the timeline so all deliverables are completed and submitted by August 22, providing a three-day buffer before the official GSoC final deadline.

10. References

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